Bayesian Credit Risk Particle Filter

TECHNICAL IMPLEMENTATION

Executive Summary

This document details a Particle Filter approach to Credit Risk Modeling with a static scenario grid. By processing debtor-specific features, the model generates comprehensive risk metrics, including **Value at Risk (VaR)**, **Expected Shortfall (ES)**, and **Economic Capital,** at the portfolio level. The model employs **Bayesian updates** to refine the Loss Distribution in real-time as default events occur. Additionally, it generates **regime probabilities**, allowing stakeholders to identify the current market state and observe shifts in the economic environment as they materialize.

Methodology

1. PARTICLE FILTER APPROACH

This implementation employs a simplified particle filtering approach adapted for credit risk assessment, where 'particles' represent discrete macroeconomic scenarios rather than evolving state trajectories. The particle grid is generated using Quasi-Monte Carlo (particularly Sobol) sequences to sample a d-dimensional space of correlated macroeconomic variables (GDP growth, unemployment rate, interest rates, inflation, and oil prices), ensuring superior low-discrepancy coverage compared to standard Monte Carlo sampling. Unlike classical particle filters that propagate particles forward via state transition models, this approach maintains a static scenario grid to enable computational efficiency through pre-computation of credit risk distributions, with evolving particle weights, updated in real-time via Bayesian likelihood functions based on observed defaults.

2. PROBABILITY OF DEFAULT (PD) GENERATION

For each particle (representing a macroeconomic state), the model generates PD for each debtor using the Logistic Regression Model for Credit Risk:

$$\alpha = \alpha_0 + \sum_i a_i \cdot Z_i + \sum_j b_j \cdot D_j$$

Where:

- D : Debtor-specific features
- Z : Macro-economic indicators for that particle (world-state scenario)
- a, b : Coefficients dependent upon the exposure class (like Real Estate) of the Debtor

The final PD is computed as:

$$PD(Z, D) = \frac{1}{1 + e^{-\alpha}}$$

3. LOSS DISTRIBUTION CONSTRUCTION

The model constructs a Loss Distribution function for each particle on the grid using the following steps:

- Borrower Profile Simulation: Generate portfolio-level losses through N iterations of a Monte Carlo Simulation.
- Uniform Random Variable Generation: For each iteration, generate uniform random variables, $U \sim \text{Uniform}(0,1)$, to simulate the stochastic nature of borrower defaults.
- Default Determination: A debtor is flagged as ‘defaulted’ when the generated uniform variable falls below their calculated Probability of Default (PD).
- Loss Aggregation: Total losses are computed by summing the Exposure at Default (EAD) for each defaulted debtor, adjusted by the recovery rates specific to their exposure class.

4. RISK ANALYSIS

All particle-specific loss distributions are aggregated together, using weights assigned to each particle. This ensures that more probable macroeconomic scenarios receive appropriate emphasis.

The aggregated loss distribution then enables calculation of standard risk measures:

- Expected Loss *(Expected value of loss distribution)*
- Unexpected Loss *(Standard Deviation of loss distribution)*
- VaR at various confidence levels *(Maximum potential loss at an α confidence level)*
- Expected Shortfall *(Expected value of loss distribution conditional on loss greater than 99% VaR)*
- Economic Capital $(99.9\%\ VaR\ -\ Expected\ Loss)$
- Tail Ratio $(99.9\%\ VaR\ /\ 99\%\ VaR)$

5. UPDATING WEIGHTS

As the model observes actual defaults, it updates particle weights using Likelihood Ratio approach:

$$\log w_t = \log w_{t-1} + \sum_{i \in D} \log PD_i + \sum_{i \notin D} \log(1 - PD_i)$$

Where:

- D : The set of unique IDs for debtors who have defaulted during the observation period.
- PD_i : The Probability of Default assigned to debtor i by that specific particle.

The updates in weights are scaled by a tempering factor (less than 1) to avoid immediate collapse of particle weights. Updated weights are then normalized, so that the total sum equals 1. By rewarding particles that explain the current credit environment, these scenarios gain more influence in the final Aggregated Loss Distribution. This allows the model to be self-correcting as more default events are observed.

6. ADVANTAGES

- This approach provides early warning detection by learning from frequent default observations rather than waiting for lagging macroeconomic statistics. Recession signals can appear weeks before official data confirms deterioration.
- It automatically adapts to changing conditions through monthly weight updates, ensuring risk measures continuously reflect the current economic outlook.
- It honestly quantifies uncertainty by maintaining probability distributions over multiple economic scenarios rather than forcing single-point regime classifications, which produces more robust tail risk estimates that account for fundamental uncertainty about which state the economy is actually in.
- It's computationally efficient - loss distributions are pre-computed once per particle and simply reweighted monthly, enabling real-time risk updates without expensive recalculation.
- This framework can be further developed to identify emerging risks within specific debtor segments, allowing for proactive mitigation strategies based on predictive insights.

7. IMPLEMENTATION CONSIDERATIONS

The model requires external calibration through other ML models. This includes:

- PD coefficients estimation through historical data
- Modeling macro-economic variables and their correlations
- Recovery rate parameters by exposure class
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Currently only debtor defaults have been considered, but this could be further developed to account for credit spreads or yield curve inversions.

A note on Particle Degeneracy and Resampling:
Unlike sequential particle filters, this approach maintains a static scenario grid, where each particle retains economic significance throughout the life of the model. The Sobol points (Quasi Monte Carlo points) generated initially are filtered out if economically unviable. Hence, particles with low weights today have the possibility of gaining weight in the future. Low probability particles often represent stress scenarios critical for regulatory

capital and risk management. Moreover, there is little computational efficiency gain, compared to the added risk of ignoring certain scenarios.

Conclusion

This approach to credit risk modeling maintains genuine uncertainty about the current economic regime while delivering actionable risk metrics for regulatory capital and portfolio management. Sequential Bayesian updates via observed default likelihoods allow the system to learn which scenarios best explain emerging reality.

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